Academic Year: 2025-26 Semester: VIII Class / Branch / Division: BE - IT A Subject: Blockchain and DLT

Home Assignment Module-6

Application of Blockchain in a Specific Domain of Your Blockchain Mini-Project

Date of Submission: 16th April 2026

Objective

The primary objective of this study is to explore how blockchain technology is transforming, or has the potential to transform, various real-world domains, with a specific focus on the gaming and digital asset domain. This assignment aims to analyze how blockchain introduces decentralization, transparency, and trust into systems that are traditionally centralized and controlled by a single authority.

The assignment also aims to evaluate the practical implementation of blockchain technology using components such as smart contracts, decentralized applications (dApps), and cryptocurrency-based transactions. It examines how these technologies can solve existing problems in traditional systems, such as lack of ownership, limited transparency, and security vulnerabilities.

Furthermore, this study seeks to analyze the feasibility of adopting blockchain solutions, considering both technical and real-world factors. It also highlights the benefits such as improved efficiency, immutability, and security, along with the challenges and limitations including scalability issues, energy consumption, regulatory concerns, and user adoption barriers.

The objective is to gain a comprehensive understanding of how blockchain can revolutionize the chosen domain and to critically assess its impact through the implementation and analysis of the NFTerra blockchain-based gaming platform.

Specific Domain Overview

The gaming and digital asset domain is a rapidly growing sector in the digital economy, where users interact within virtual environments and engage in activities such as playing games, earning rewards, and collecting virtual items. These digital assets may include characters, skins, weapons, virtual land, and other in-game items.

In traditional gaming systems, these assets are controlled and managed by centralized game developers or companies. Players can use these assets within the game but do not have actual ownership over them.

In the context of the NFTerra project, this domain is enhanced using blockchain technology by introducing Dynamic NFTs (Non-Fungible Tokens). These NFTs represent unique digital assets that can evolve over time based on user actions, such as leveling up from one stage to another (e.g., Egg to Creature to Dragon to Phoenix to Immortal). This creates a more interactive and engaging gaming experience while also enabling real ownership of digital assets.

Key Challenges:

1. Centralized control by game developers
2. Limited interoperability between different gaming platforms
3. Risk of fraud, hacking, and asset duplication
4. No real-world value for virtual in-game items
5. Lack of transparency in reward and transaction systems

6. Dependence on centralized servers for data storage
7. Possibility of account bans leading to loss of assets
8. Limited control of players over in-game economy
9. High chances of data manipulation by authorities

Blockchain Application in the Domain

1. Addressing Challenges in the Domain using Blockchain

Blockchain technology plays a significant role in addressing the challenges present in the traditional gaming domain. It enables true ownership of digital assets through Non-Fungible Tokens (NFTs), allowing players to have complete control over their in-game items.

By eliminating centralized control, blockchain ensures decentralization, where no single authority can manipulate data or restrict access. The use of immutable ledgers prevents fraud, duplication, and unauthorized modifications, thereby increasing trust and transparency.

How blockchain addresses the challenges in the domain:

1. Provides true ownership of in-game assets through NFTs
2. Eliminates centralized authority using decentralization
3. Ensures transparency via public blockchain records
4. Prevents fraud and duplication using immutability
5. Enables secure peer-to-peer transactions without intermediaries

The implementation of blockchain in the NFTerra project is achieved using smart contracts developed in Solidity and deployed on the Ethereum. These smart contracts automate processes such as NFT minting, ownership transfer, and level-based evolution of dynamic NFTs. Decentralized storage is used to securely store NFT metadata, ensuring data availability and protection against tampering or loss.

Several real-world platforms demonstrate the effectiveness of blockchain in the gaming industry. For example, Axie Infinity allows users to earn and trade NFT-based assets, while Decentraland enables ownership of virtual land in a decentralized environment. Similarly, The Sandbox and Gods Unchained showcase how blockchain technology can create secure and economically valuable gaming ecosystems.

2. Use of smart contracts, decentralized storage, or tokenization (as applicable).

1. Smart Contract

In the NFTerra project, a smart contract is implemented to manage NFT creation and evolution. The contract includes functions for minting NFTs and upgrading their levels, along with pricing mechanisms for each action.

In the NFTerra project, a smart contract is implemented to manage the complete lifecycle of NFTs, including their creation, ownership, and evolution. The smart contract acts as the core logic of the system, ensuring that all operations are executed in a secure, transparent, and automated manner without the need for intermediaries.

The contract maintains important state variables such as the total number of NFTs (tokenCounter), minting cost (mintPrice), level-up cost (levelUpPrice), and a mapping that tracks the level of each NFT. These variables help in controlling the behavior of the system and maintaining consistency across all transactions.

Structure of Smart Contract: DynamicNFT

Variables:

- tokenCounter
- mintPrice (0.05 ETH)
- levelUpPrice (0.02 ETH)
- levels mapping (tokenId → level)

Functions:

- mintNFT(tokenURI)
- levelUp(tokenId)
- ownerOf(tokenId)

The mintNFT() function allows users to create new NFTs by paying a fixed amount of 0.05 ETH, while the levelUp() function enables users to upgrade their NFTs by paying 0.02 ETH. Each NFT is assigned a level that increases over time, making it a dynamic NFT. Ownership is securely managed using blockchain.

2. Storage

In the NFTerra project, NFT metadata is currently stored **on-chain** using the ERC721URIStorage standard. The metadata is encoded in **base64 format** and stored as a data URI (data:application/json; base64,...) within the smart contract itself. This approach ensures that important information such as NFT name.

However, the image assets associated with the NFTs are stored on a **centralized server** (API-based image hosting). This means that while the metadata is secure and decentralized to some extent, the visual representation of NFTs depends on a centralized system, which introduces a potential risk of data unavailability if the server goes offline.

At present, the project does not use fully decentralized storage solutions such as IPFS or Arweave. These technologies are commonly used in blockchain applications to store large files like images in a distributed manner. Their absence means that the system is only partially decentralized.

3. Tokenization

In the NFTerra project, tokenization is implemented by converting in-game digital assets into Non-Fungible Tokens (NFTs). Each NFT represents a unique digital asset with its own token ID, owner address, level, and metadata. These tokens are created using the smart contract and are stored on the blockchain, ensuring secure and verifiable ownership.

Each tokenized asset can be upgraded through levels, making it dynamic in nature. Additionally, these NFTs can be bought, sold, and traded in the marketplace, giving them real-world value.

Tokenized NFT

- Token ID
- Owner Address
- Level
- Rarity
- Metadata (tokenURI)

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Each NFT in the system can evolve through different levels, making it dynamic in nature. Additionally, these tokenized assets can be bought, sold, and traded in the marketplace, giving them real-world value and enabling a decentralized digital economy.

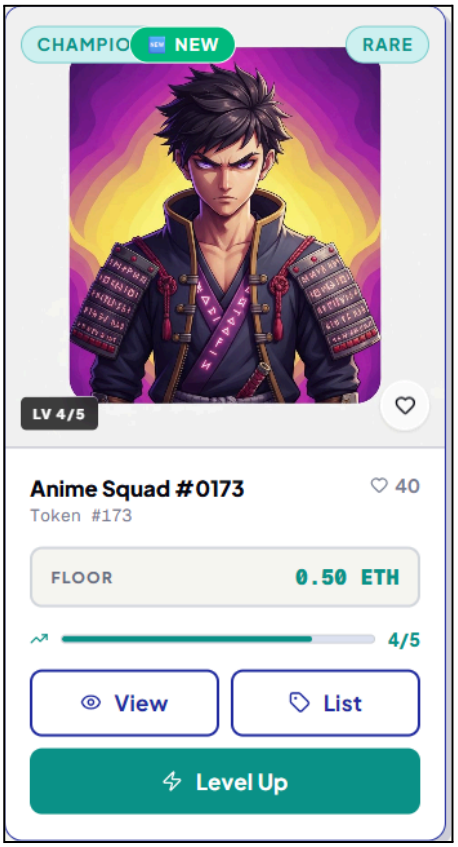

Figure: Example of Tokenized Dynamic NFT in NFTerra Platform

The above figure shows a tokenized NFT from the NFTerra platform. Each NFT has a unique token ID, level, rarity, and ownership details. The level indicator (Level 4/5) represents the dynamic evolution of the NFT. The price displayed in ETH highlights its real-world value, while the options such as "View", "List", and "Level Up" demonstrate the ability to interact, trade, and upgrade the asset.

3. Existing solutions or case studies (e.g., real-world companies using blockchain in this domain).

Several real-world platforms demonstrate the successful application of blockchain technology in the gaming and digital asset domain. These platforms utilize blockchain to enable ownership, trading, and monetization of digital assets through NFTs.

Platform	Description	Use of Blockchain	Key Features
Axie Infinity	Play-to-earn blockchain game	NFTs represent characters (Axies)	Breeding, battling, trading NFTs
Decentral and	Virtual world with digital land	Land is owned as NFTs	Buy, sell, build on virtual land
The Sandbox	Metaverse gaming platform	Assets and land tokenized as NFTs	User-created games, monetization
OpenSea	NFT marketplace platform	Enables buying/selling of NFTs	Supports multiple NFT collections and assets

These case studies highlight how blockchain technology enables transparency, ownership, and decentralized economies in real-world gaming applications.

Benefits of Using Blockchain in Gaming Domain

1. **Transparency:** Blockchain provides a transparent system where all transactions and ownership records are publicly available and can be verified by any user, increasing trust and ensuring fairness.
2. **Immutability:** Data stored on the blockchain cannot be modified or deleted, ensuring that ownership and transaction history remain permanent and tamper-proof.
3. **Security:** Blockchain uses cryptographic techniques to secure transactions and digital assets, reducing the risk of fraud and unauthorized access.
4. **Efficiency:** Smart contracts automate processes such as minting and trading, reducing intermediaries and improving system efficiency.

Challenges and Limitations

1. **Scalability:** Blockchain networks may experience slower performance and higher costs as the number of users and transactions increases.
2. **Energy Consumption:** Some blockchain systems require high computational power, leading to increased energy usage.
3. **Regulatory Issues:** Legal frameworks for blockchain and NFTs are still developing, creating uncertainty.
4. **User Adoption:** Lack of awareness and technical complexity make it difficult for new users to adopt blockchain systems.

Text Book:

1. Blockchain Technology, Chandramouli Subramanian, Asha A George, Abhillash K. A and Meena Karthikeyen,

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